

Bhumi Godiwala

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EDUCATION

- Arizona State University | Tempe, AZ**
PhD in Computer Science (GPA 4.0/4.0)
- Jan 2026 – Present
- University of Southern California | Los Angeles, CA**
Master of Science in Computer Engineering — Machine Learning & Data Science specialization (GPA 3.8/4.0)
- August 2021 – May 2023
- D.J. Sanghvi College of Engineering (Univ. of Mumbai) | Mumbai, India**
Bachelor of Engineering in Electronics & Telecommunications Engineering (CGPA 9.19/10)
- Jul 2016 – Oct 2020

PROFESSIONAL EXPERIENCE

- ASU Decision Theater Network | Tempe, AZ**
Graduate Research Data Scientist
- Apr 2026 – Present
- Architect end-to-end deep learning deployment and inference pipelines across multiple research domains, including a Valley Fever disease detection model, taking prototypes to production-ready model serving.
 - Build an NLP-based skill-extraction system mapping university course offerings to job market requirements, surfacing actionable skill gaps at scale.
 - Lead clustering analysis on Commonwealth Fund health indicators for Maricopa County, segmenting regional health-outcome patterns for a county-level health-impact research initiative.
 - Co-authoring a research paper on LLM and RAG-based approaches combined with topic modeling for large-scale text analysis.
- Infosys Ltd | Houston, TX**
Senior Data Scientist
- Jan 2025 – Nov 2025
- Engineered an ML anomaly-detection system boosting utility theft detection accuracy 25%, improving revenue protection and lead conversion for a regulated energy utility serving 1M+ customers.
 - Redesigned Budget Billing analytics to surface behavioral drivers, increasing enrollment 15% and retention 10%; deployed findings as policy with engineering and product teams.
- MemoryCare AI (now Scienza Health) | Newport Beach, CA**
Sr. AI/ML Engineer
- May 2024 – Jan 2025
- Built LLM + RAG pipelines (Python, TensorFlow) improving retrieval accuracy and powering HIPAA-compliant patient assistance.
 - Created an avatar-based speech and behavior tracking system (PyTorch, OpenCV) achieving 90% detection accuracy for early Alzheimer's/TBI intervention.
 - Designed an Azure-based cognitive assessment platform, cutting test time 35% while scaling securely across clinics.
- USC Information Sciences Institute | Marina Del Rey, CA**
Machine Learning Engineer
- Jul 2023 – Jul 2024
- Developed Adaptive Mixture Quantization (AMQ) for Cloud/Edge AI, improving accuracy 5%, reducing model size 15%, and boosting comms efficiency 30%.
- TetraMem Inc | Fremont, CA**
Machine Learning Intern
- May 2022 – May 2023
- Optimized Human Pose Estimation and Visual Wake Words models for edge inference chips, achieving 93% accuracy with reduced memory and latency.
 - Applied Quantization Aware Training and post-training quantization, preserving accuracy while shrinking model parameters.
- Tata Consultancy Services | Mumbai, India**
Software Engineer
- October 2020 – August 2021
- Developed and deployed full-stack Java/HTML/CSS/JS modules for university portals serving 10,000+ students; automated metadata mapping and reporting, cutting manual processing time 30%.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, Git, Bash
ML/DL: PyTorch, TensorFlow, Scikit-Learn, Keras, HuggingFace Transformers, ONNX, OpenCV, QAT
GenAI/LLM: RAG Pipelines, LLM Fine-tuning, Prompt Engineering, GPT-3/4, Embedding Models, Topic Modeling
Cloud/MLOps: AWS (S3, Lambda, EC2), Oracle Cloud, Azure Cognitive Services, Spark, Hadoop, Docker

ACADEMIC PROJECTS

- LLM-Powered Email Generation System**
Built an end-to-end LLM email automation system with a FastAPI + React frontend, deployed on AWS for scalable, serverless, real-time generation.
Technologies: Python, FastAPI, React, Docker, AWS (Lambda, EC2, API Gateway)
- ASL Gesture Prediction using ST-GAN for Shadow Removal**
Engineered a GAN-CNN fusion model to eliminate shadow noise and improve ASL classification accuracy, achieving 92.9% test accuracy with full MLflow experiment tracking.
Technologies: Python, PyTorch, TensorFlow, GAN, CNN, MLflow